

Tianjian Li

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Research Interests

My research lies at the intersection of Natural Language Processing and Machine Learning. I focus on **data engineering** for language models (LMs), aiming to develop **online and adaptive data methods** that adjust dynamically to the model's training progress — including [online filtering of low-quality data](#), [adaptive data balancing across sources](#), and [enhancing data diversity in online Reinforcement Learning](#).

Education

Johns Hopkins University, Ph.D. in Computer Science Aug 2024 – Present

- Advisor: Daniel Khashabi

Johns Hopkins University, M.S. in Computer Science Aug 2022 - May 2024

- Advisor(s): Kenton Murray, Philipp Koehn, Daniel Khashabi

New York University, B.A. Joint Major in Mathematics and Computer Science Aug 2017 - May 2021

Industry Experience

Research Scientist Intern, Meta Fundamental AI Research (FAIR) – New York, NY June 2026 – Sept 2026

- Host: Jason Weston.

Research Scientist Intern, Meta Fundamental AI Research (FAIR) – Bellevue, WA May 2025 – Dec 2025

- Host: Tianlu Wang; Manager(s): Jack Lanchantin, Jason Weston.

Machine Learning Engineer, Baidu Inc. – Beijing, China Aug 2021 – Jan 2022

First-Authored Publications

Jointly Reinforcing Diversity and Quality in Language Model Generations [\[Paper\]](#) [\[Code\]](#)

Tianjian Li, Yiming Zhang, Ping Yu, Swarnadeep Saha, Daniel Khashabi, Jason Weston, Jack Lanchantin, Tianlu Wang.

Scaling Post-training (SPOT) workshop at ICLR 2026.

Reasoning over Mathematical Objects: On-Policy Reward Modeling and Test Time Aggregation [\[Paper\]](#)

Pranjal Aggarwal, Marjan Ghazvininejad, Seungone Kim, Ilia Kulikov, Jack Lanchantin, Xian Li, **Tianjian Li***, Bo Liu, Graham Neubig, Anaelia Ovalle, Swarnadeep Saha, Sainbayar Sukhbaatar, Sean Welleck, Jason Weston, Chenxi Whitehouse, Adina Williams, Jing Xu, Ping Yu, Weizhe Yuan, Jingyu Zhang, Wenting Zhao. Preprint 2026.

*Alphabetical order. Lead the section on parallel test-time scaling.

SimpleMix: Frustratingly Simple Mixing of Off- and On-policy Data in Language Model Preference Learning [\[Paper\]](#)

Tianjian Li, Daniel Khashabi.

ICML 2025.

Upsample or Upweight: Balanced Training on Heavily Unbalanced Datasets [\[Paper\]](#)[\[Video\]](#)

Tianjian Li, Weiting Tan, Haoran Xu, Kenton Murray, Daniel Khashabi.

NAACL 2025.

Error Norm Truncation: Robust Training in the Presece of Data Noise in Text Generation Models [\[Paper\]](#)[\[Code\]](#)[\[Video\]](#)

Tianjian Li, Haoran Xu, Philipp Koehn, Daniel Khashabi, Kenton Murray.

ICLR 2024. [Spotlight Presentation \(367/7304 \$\approx\$ 5%\)](#).

Why Does Zero-shot Cross-lingual Transfer Fail? An Explanation and a Solution

[\[Paper\]](#)

Tianjian Li, Kenton Murray.

ACL 2023 Findings.

Non-First Authored Publications

Many-Tier Instruction Hierarchy in LLM Agents

[\[Paper\]](#)

Jingyu Zhang, Tianjian Li, William Jurayj, Hongyuan Zhan, Benjamin Van Durme, Daniel Khashabi. Preprint 2026.

The Flaw of Averages: Quantifying Uniformity of Performance on Benchmarks

[\[Paper\]](#)

Arda Uzunoglu, Tianjian Li, Daniel Khashabi. Preprint 2025.

The Translation Barrier Hypothesis: Multilingual Generation with Large Language Models Suffers from Implicit Translation Failure

[\[Paper\]](#)

Niyati Bafna, Tianjian Li, Kenton Murray, David R. Mortensen, David Yarowsky, Hale Sirin, Daniel Khashabi. AACL 2025.

Benchmarking Language Model Creativity: A Case Study on Code Generation

[\[Paper\]](#)[\[Code\]](#)[\[Video\]](#)

Yining Lu, Dixuan Wang, Tianjian Li, Dongwei Jiang, Sanjeev Khudanpur, Meng Jiang, Daniel Khashabi. NAACL 2025.

Verifiable by Design: Aligning Language Models to Quote from Pre-Training Data

[\[Paper\]](#)[\[Code\]](#)[\[Video\]](#)

Jingyu Zhang, Marc Marone, Tianjian Li, Benjamin Van Durme, Daniel Khashabi. NAACL 2025. [Oral Presentation.](#)

Technologies

Languages: Python, C++ , Java, Shell Script.

Frameworks and Tools: Linux, PyTorch, HuggingFace, Inference Engines (vLLM, SgLang), RL frameworks (verl, slime), Distributed Training (FSDP, DeepSpeed), vim, Latex.

Awards

National Olympiad in Informatics Provinces (NOIP) First Prize.

2015

Invited Talks

Beijing Academy of Artificial Intelligence (BAAI)

Dec 2025

Johns Hopkins University, CLSP Seminar

Spring 2024

Service

Reviewer for —

- NLP Conferences: ACL (2023, 2024, 2025), EMNLP (2023, 2024, 2025), NAACL (2025), EACL (2024, 2025), COLM (2024, 2025, 2026).
- ML Conferences: ICLR (2025, 2026), NeurIPS (2025), ICML (2026).

Organizer: Mid-Atlantic Student Colloquium on Speech, Language and Learning (2024)

Teaching

Head Teaching Assistant, JHU CS 601.471/671 NLP: Self-supervised Models

Spring 2025

- Gave lectures on distributed training: [\[Video 1\]](#)[\[Video 2\]](#)

Course Assistant, JHU CS 601.471/671 NLP: Self-supervised Models

Spring 2024